

Interleaving for time-varying fading channels

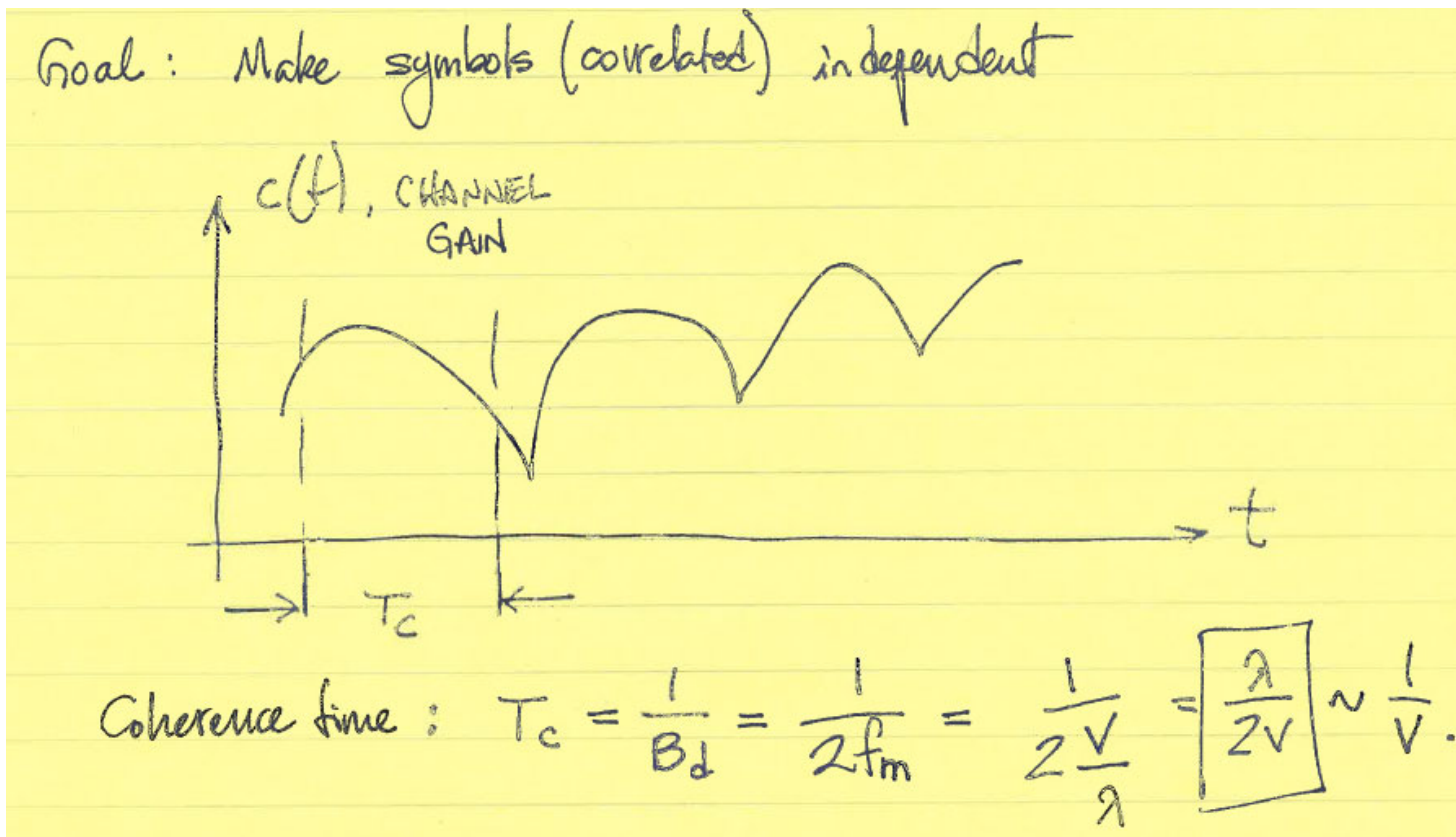
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EE161: Digital Communication Systems

San Jose State University

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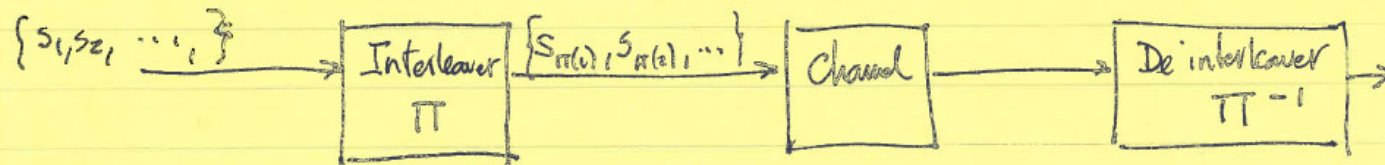
Coherence and correlation between symbols



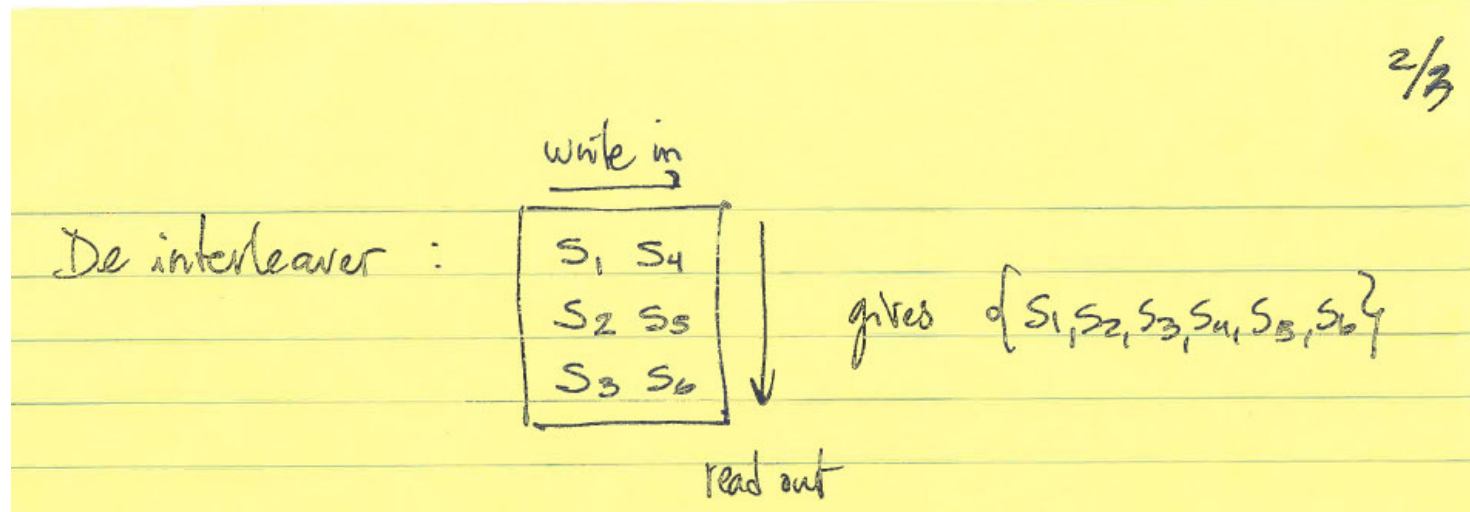
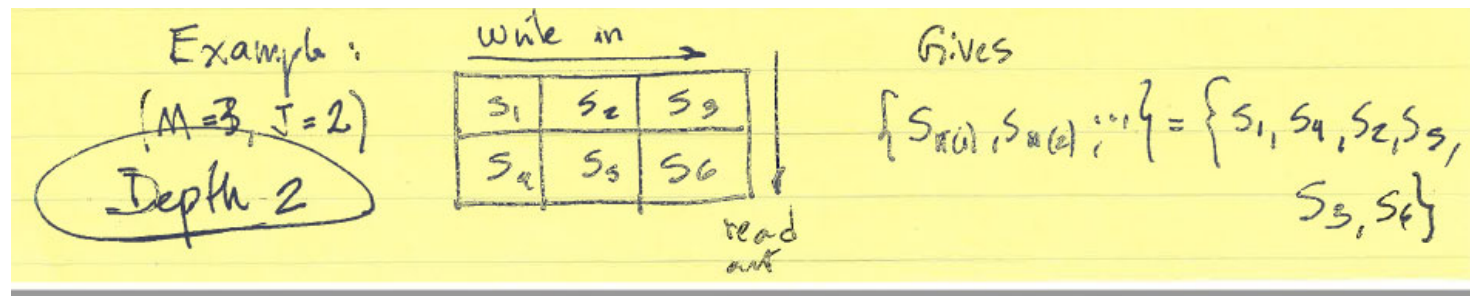
Idea: Scramble modulated symbols

⇒ "Scramble" correlated symbols (over T_c seconds)
so that the received sequence has statistically
independent symbols.

Interleaver (block type. there are other types)



Example: Two-by-three block interleaver



General J -by- M block interleaver:

(Symbol indices shown)

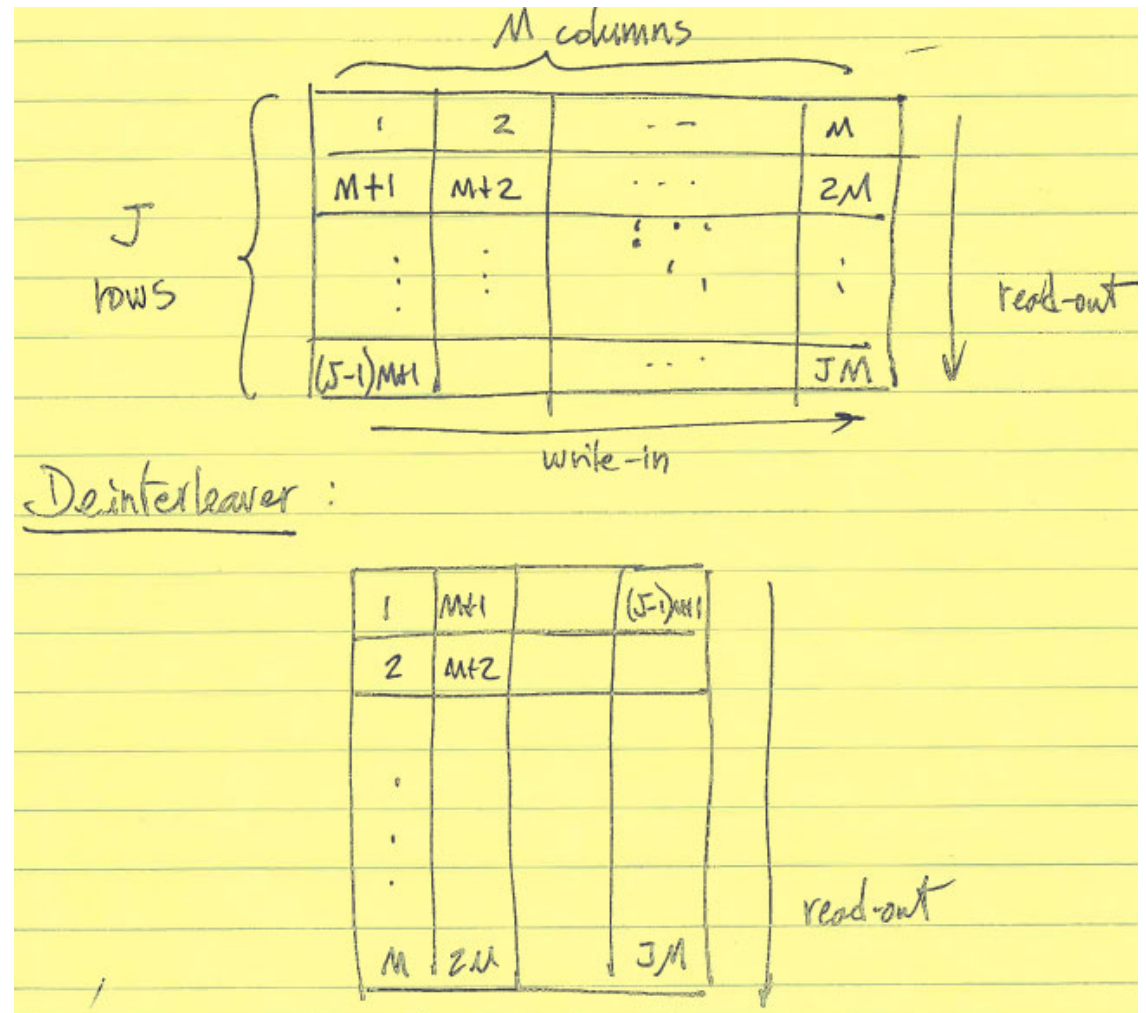
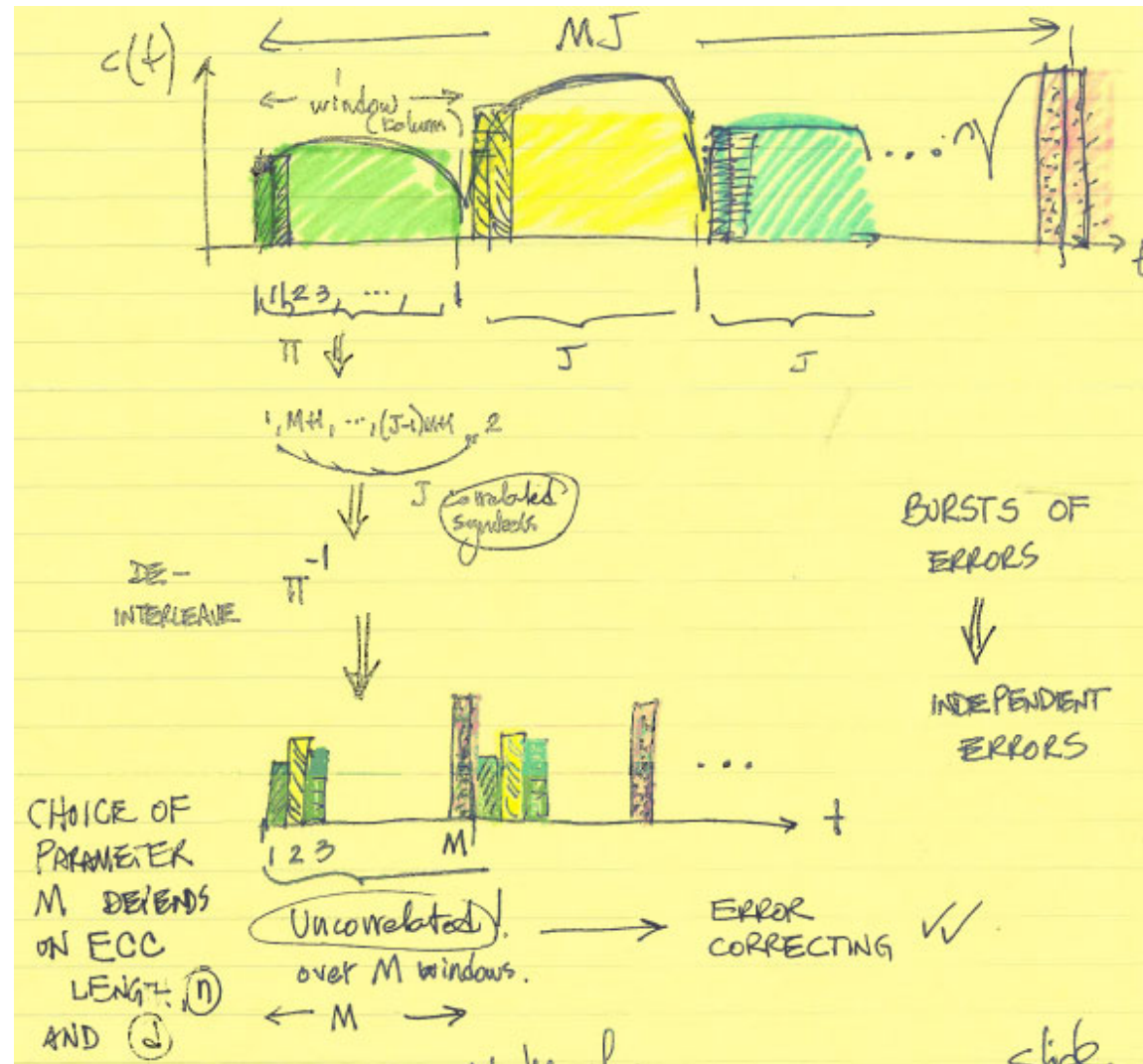
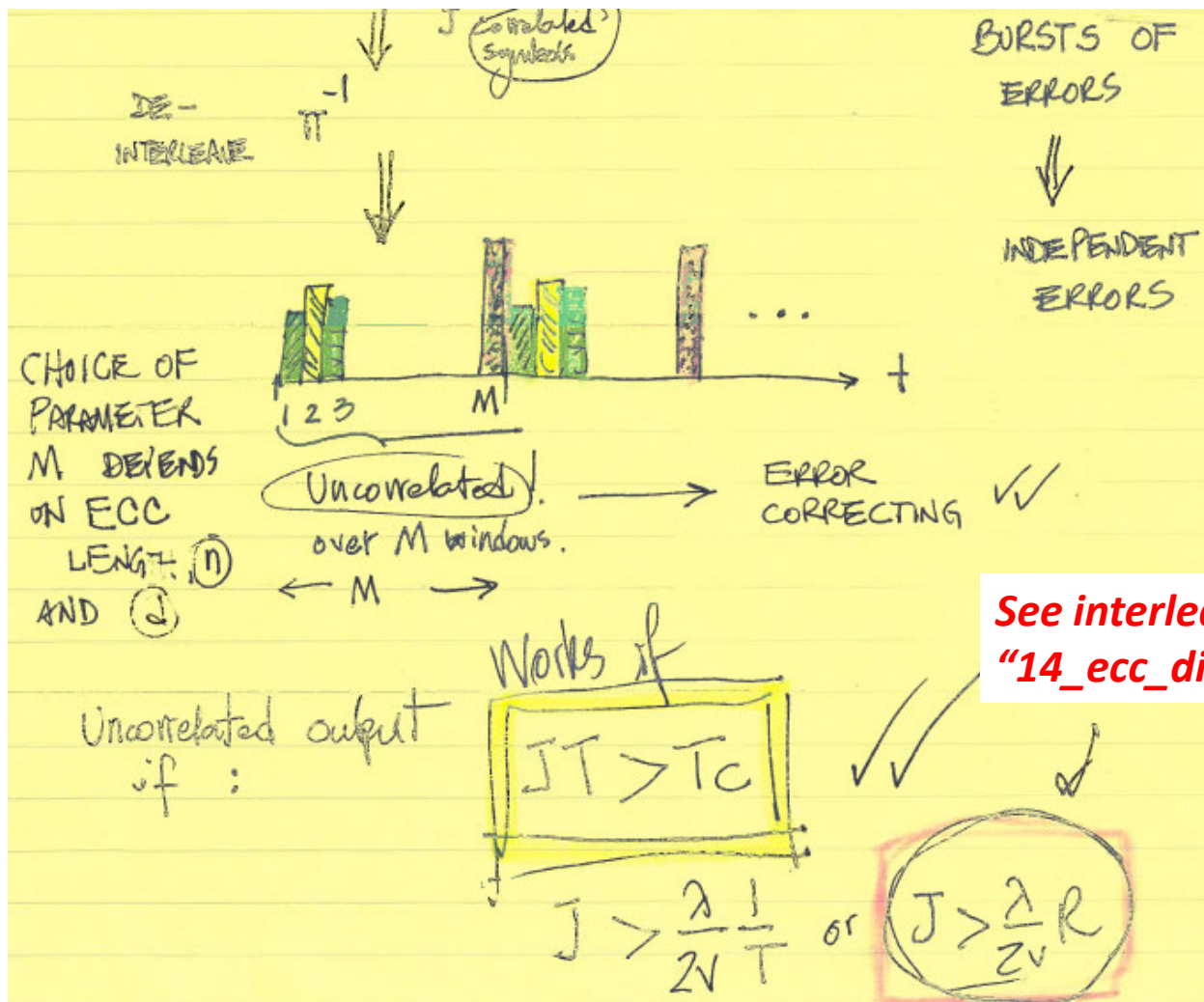


Illustration:





See interleaver note: "14_ecc_diversity_note.pdf"

Performance of ECC. SPC (3,2,2) example

